

AI in Finance

Arpit Gupta (NYU Stern)
Spring 2026
Session 5

Fraud Detection and Compliance



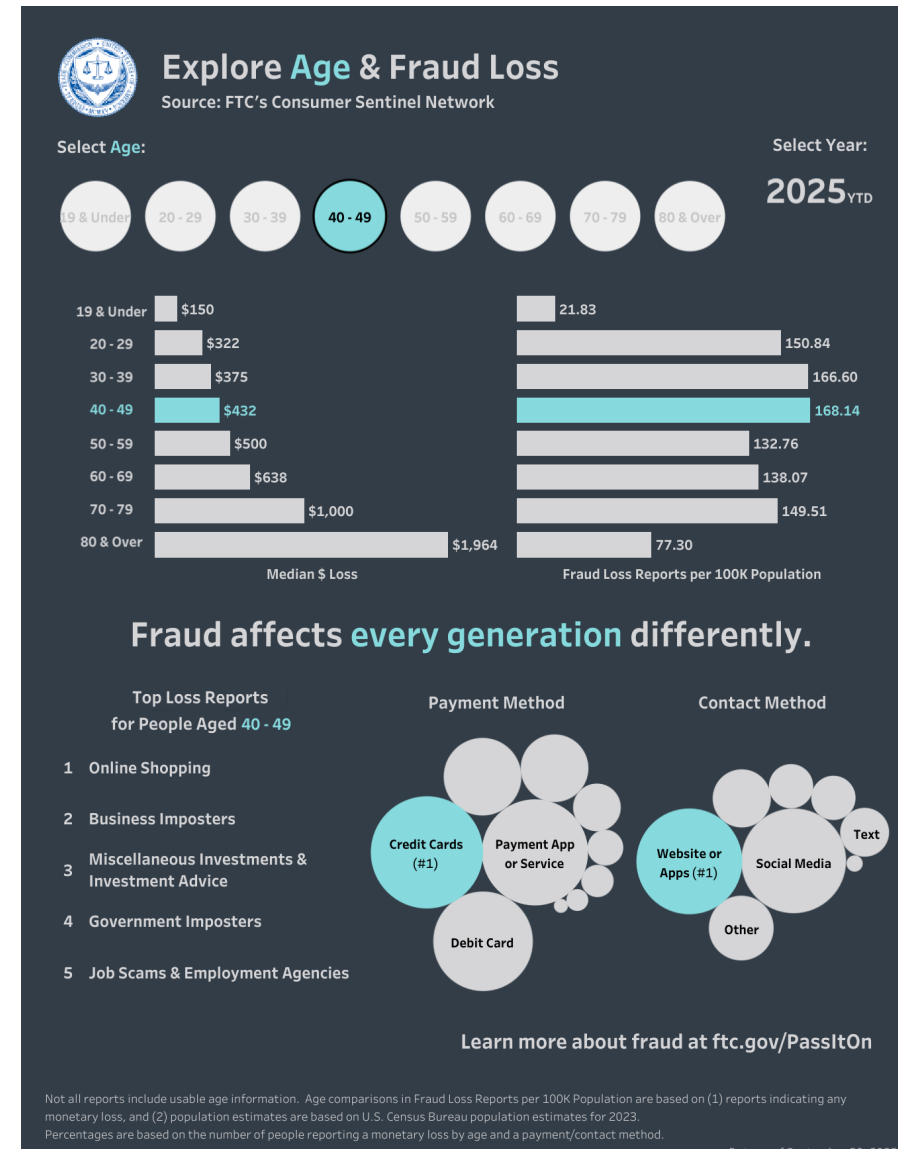
Background



The Cardsharps (Caravaggio)

Fraud is Pervasive

- Consumers report (to FTC) losing \$12.5b in fraud in 2024
- Global payment card fraud roughly \$30b in 2024, but likely underestimated
- IMF and World Bank estimate \$2.2-\$3.6 trillion is subject to money-laundering every year
 - Evading sanctions, capital controls, financing crime and terrorism

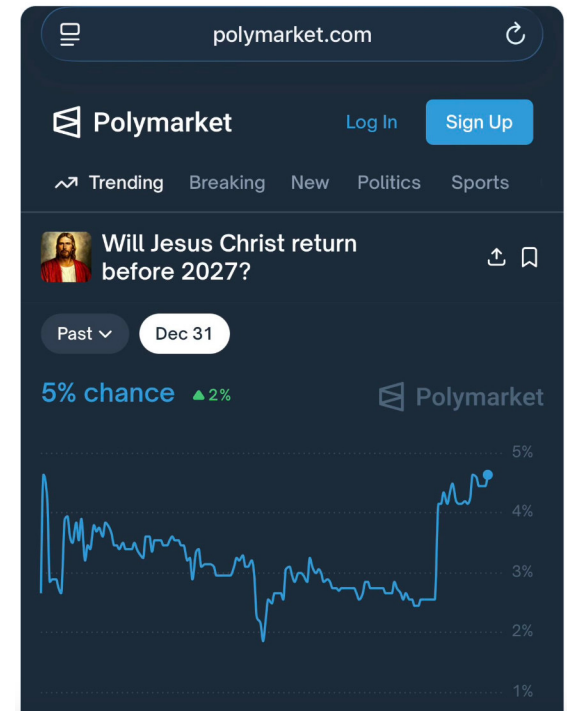


(Non-Comprehensive) List of Financial Crimes

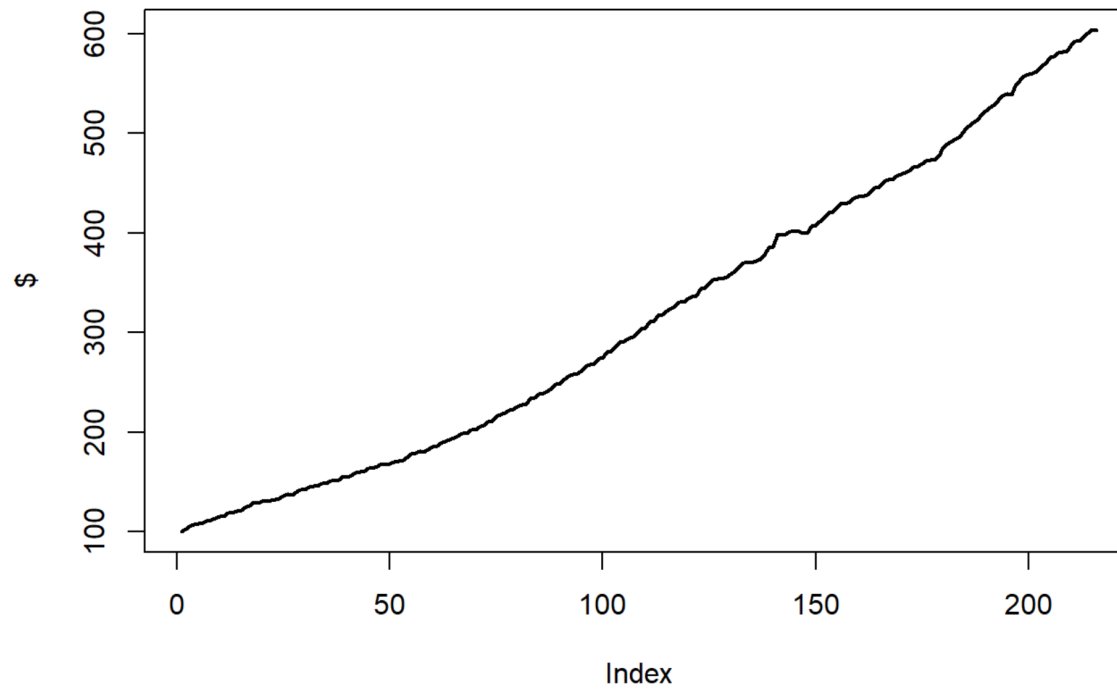
- Credit card fraud
- Online payment fraud
- Insurance fraud
- Money laundering
- Identity theft/phishing
- Mortgage fraud
- Insider Trading
- Market Manipulation (right ->)
- Embezzlement
- Tax evasion



So the reason this Polymarket "prediction market" is performing so insanely high is because there's a second market asking if this market will go above 5%. People in the derivative market are manipulating this market. Which defeats the public policy case for prediction markets...



What's the Problem Here?



KYC (Know Your Customer)

- Key part of how financial institutions verify consumer identity
- **Customer Identification Program (CIP):**
 - For individuals: personally verifiable information
 - Corporations: corporate structure and beneficial (ultimate) ownership
- **Customer Due Diligence (CDD):** Expected pattern of transactions and behaviors for customer profile
 - Enhanced Due Diligence (EDD) for politically exposed persons (PEP)

Anti-Money Laundering (AML)

- Regulatory framework to detect and prevent the movement of illicit funds
- Verify customers at point of onboarding (KYC)
- Transactions monitoring: tagging unusual patterns like spikes in volume, structuring transactions to fall below reporting thresholds, activity inconsistent with profile etc.
 - Sanctions screens and Suspicious Activities Reports, filed with FinCEN
- Global spending on KYC and AML is ~\$3b in 2025

Finding Needles in Haystacks

The Confusion Matrix of Fraud Detection

Every fraud system makes two kinds of mistakes — and they have very different costs

	ACTUALLY Fraud	ACTUALLY Legitimate
MODEL SAYS Fraud	✓ True Positive Correctly caught fraud. The system works as intended. 🎯 Goal: maximize these	⚠️ False Positive — Type I Error Legitimate transaction blocked. Customer frustrated, revenue lost, analyst time wasted. 💰 Cost: ~\$50–150 per alert to investigate
MODEL SAYS Legitimate	✗ False Negative — Type II Error Fraud gets through undetected. Direct financial loss, regulatory penalties, reputational damage. 🔥 Cost: avg \$4.88M per data breach (IBM 2024)	✓ True Negative Legitimate transaction approved. Smooth customer experience. ✅ ~99.9% of all transactions

THE CORE TRADEOFF

Tightening rules to catch more fraud (↓ Type II) inevitably blocks more legitimate transactions (↑ Type I). Traditional rule-based systems have **95%+ false positive rates** — meaning for every real fraud caught, ~19 legitimate customers are flagged. ML models improve this ratio, but the asymmetry in costs means the optimal threshold depends on your business model: a payments company losing customers values low Type I; a bank facing regulatory scrutiny prioritizes low Type II.

What Does This Mean for Fraud Detection?

- More “accurate” fraud detection algorithm is always classify “never fraud”
 - This will be 99+% accurate! High precision (no false positives), poor recall
 - But this would be a very bad algorithm in practice (why?)
- Because the “base rate” of fraud is so low, any fraud algorithm is going to have a lot of false positives
 - These are costly to address, annoy customers, etc.
- The cost of lowering false positives is increasing false negatives
 - Allowing genuine fraud is typically even more costly

High False Positive Rates Per True Fraud

RESEARCH ARTICLE | OCTOBER 01 2022

The Cost of Fraud Prediction Errors

Messod D. Beneish ; Patrick Vorst

[+ Author & Article Information](#)

The Accounting Review (2022) 97 (6): 91–121.

<https://doi.org/10.2308/TAR-2020-0068>

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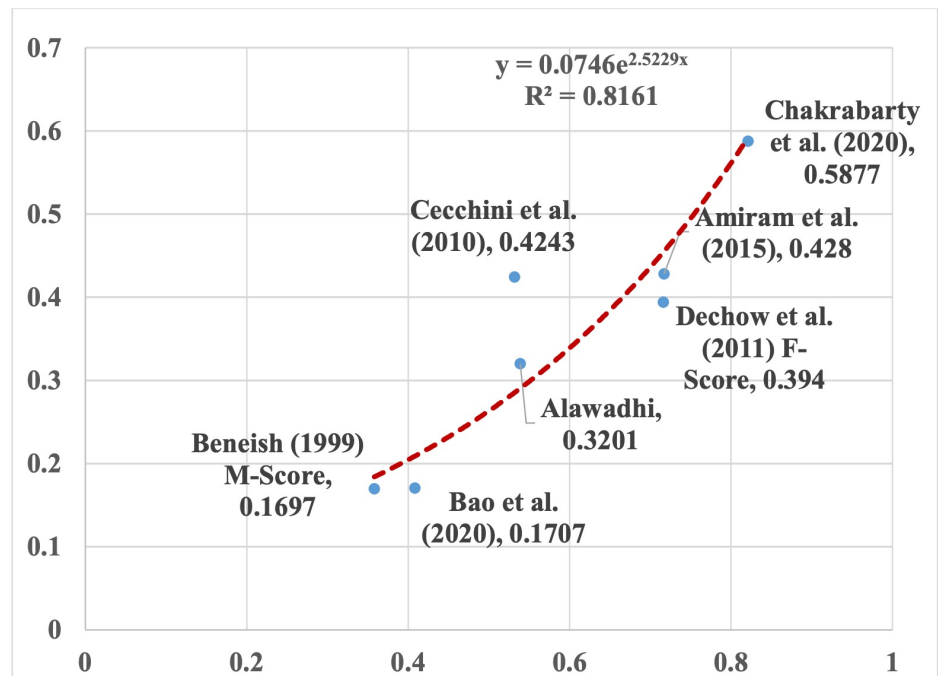
 Tools

 Cite

ABSTRACT

We compare seven fraud prediction models with a cost-based measure that nets the benefits of correctly anticipating instances of fraud against the costs borne by incorrectly flagging non-fraud firms. We find that even the best models trade off false to true positives at rates exceeding 100:1. Indeed, the high number of false positives makes all seven models considered too costly for auditors to implement, even in subsamples where misreporting is more likely. For investors, M-Score and, at higher cut-offs, the F-Score, are the only models providing a net benefit. For regulators, several models are economically viable as false positive costs are limited by the number of investigations regulators can initiate, and by the relatively low market value loss a “falsely accused” firm would bear in denials of requests under the Freedom of Information Act (FOIA). Our results are similar whether we consider fraud or two alternative restatement samples.

Figure 3: False Positive Rates versus True Positive Rates



False positive rates are depicted on the vertical axis, true positive rates on the horizontal axis.

AI for Fraud Detection

Review of Fraud Detection Techniques

- Pre-1990s: Manual Review
 - Slow, expensive, doesn't scale
- 1990s-2000s: Rule-Based Systems. I.e. flag transactions if >\$10k
 - Scalable but rigid, easy to game, high false positives (>95%)
- 2010s: Machine Learning. Supervised models learn fraud from labeled data
 - Better accuracy, can adapt over time to new patterns
- 2015-2020s: Deep Learning. Neural nets, graph networks to map transaction relationships
 - Captures novel frauds, but “black box”
- 2020s: Gen AI/LLMs analyze unstructured data (narratives)
 - Can also be used for drafting regulatory document review

Bag of Words Fraud Detection

Accounting Variables, Deception, and a Bag of Words: Assessing the Tools of Fraud Detection*

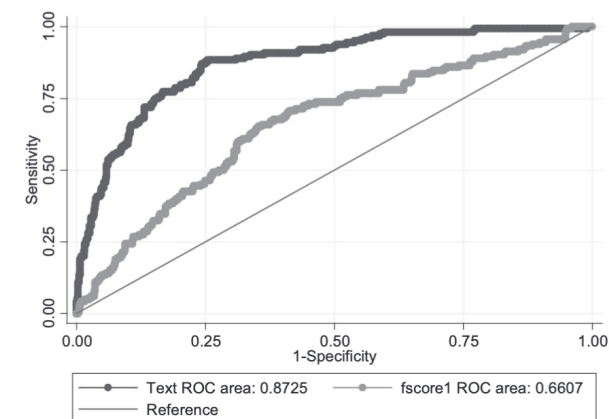
LYNNETTE PURDA, *Queen's University*

DAVID SKILLICORN, *Queen's University*

Abstract

We develop a data-generated tool for distinguishing between fraudulent and truthful reports based on the language used in the management discussion and analysis section of annual and interim reports. Using this method, we are able to assign a probability of truth to each report which is then shown to be an effective indicator of fraud. Our work goes beyond the development of a tool alone, however, by conducting an extensive comparison of our probability-of-truth measure with eight alternative detection tools representing both quantitative and language-based approaches. Comparisons are made across a variety of samples and show that our language-based approach can be effective in both cross-sectional and time-series settings. It is useful both in distinguishing between fraudulent and truthful firms and in identifying fraudulent reports from a series of reports issued by a single firm. This second setting is one in which accounting-based detection tools have frequently struggled. We establish that, not only is our probability-of-truth measure significantly associated with fraud, so too is the *change* in this measure from a firm's previous reports. Prior reports may serve an important benchmarking role in using language-based tools to identify fraud.

Figure 1 Area under the ROC curve: Probability of truth versus *F*-score



Description:

The figure plots the area under the Receiver-Operating-Characteristic (ROC) curve for our text-based probability-of-truth measure and Dechow et al.'s (2011) *F*-score (Model 1). The two fraud indicators are applied to a sample of 802 annual reports from firms with at least one AAER bulletin asserting fraud. ROC curves plot the true positive rate (sensitivity) versus the false-positive rate (1-Specificity). An area under the curve greater than 0.50 implies that the fraud indicator has some ability to correctly classify fraudulent and truthful reports with larger numbers corresponding to improved classification. A value of 0.50 implies that the indicator has no more power than a random classifier.

Credit Card Fraud Detection

Credit Card Fraud Detection: A Realistic Modeling and a Novel Learning Strategy

Andrea Dal Pozzolo, Giacomo Boracchi, Olivier Caelen, Cesare Alippi, *Fellow, IEEE*,
and Gianluca Bontempi, *Senior Member, IEEE*

Abstract—Detecting frauds in credit card transactions is perhaps one of the best testbeds for computational intelligence algorithms. In fact, this problem involves a number of relevant challenges, namely: concept drift (customers’ habits evolve and fraudsters change their strategies over time), class imbalance (genuine transactions far outnumber frauds), and verification latency (only a small set of transactions are timely checked by investigators). However, the vast majority of learning algorithms that have been proposed for fraud detection rely on assumptions that hardly hold in a real-world fraud-detection system (FDS). This lack of realism concerns two main aspects: 1) the way and timing with which supervised information is provided and 2) the measures used to assess fraud-detection performance. This paper has three major contributions. First, we propose, with the help of our industrial partner, a formalization of the fraud-detection problem that realistically describes the operating conditions of FDSs that everyday analyze massive streams of credit card transactions. We also illustrate the most appropriate performance measures to be used for fraud-detection purposes. Second, we design and assess a novel learning strategy that effectively addresses class imbalance, concept drift, and verification latency. Third, in our experiments, we demonstrate the impact of class unbalance and concept drift in a real-world data stream containing more than 75 million transactions, authorized over a time window of three years.

Index Terms—Concept drift, Credit card fraud detection, learning in nonstationary environments, unbalanced classification.

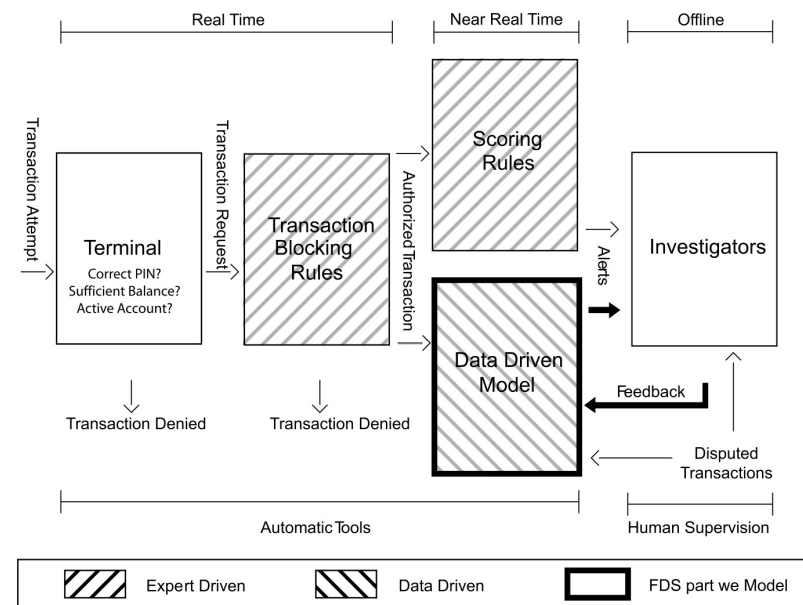


Fig. 1. Scheme illustrating the layers of control in an FDS. Our focus is mainly on the DDM and the alert-feedback interaction, which regulates the way recent supervised samples are provided.

Embeddings

DeepTrax: Embedding Graphs of Financial Transactions

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Abstract—Financial transactions can be considered edges in a heterogeneous graph between entities sending money and entities receiving money. For financial institutions, such a graph is likely large (with millions or billions of edges) while also sparsely connected. It becomes challenging to apply machine learning to such large and sparse graphs. Graph representation learning seeks to embed the nodes of a graph into a euclidean vector space such that graph topological properties are preserved after the transformation. In this paper, we present a novel application of representation learning to bipartite graphs of credit card transactions in order to learn embeddings of account and merchant entities. Our framework is inspired by popular approaches in graph embeddings and is trained on two internal transaction datasets. This approach yields highly effective embeddings, as quantified by link prediction AUC and F1 score. Further, the resulting entity vectors retain intuitive semantic similarity that is explored through visualizations and other qualitative analyses. Finally, we show how these embeddings can be used as features in downstream machine learning business applications such as fraud detection.

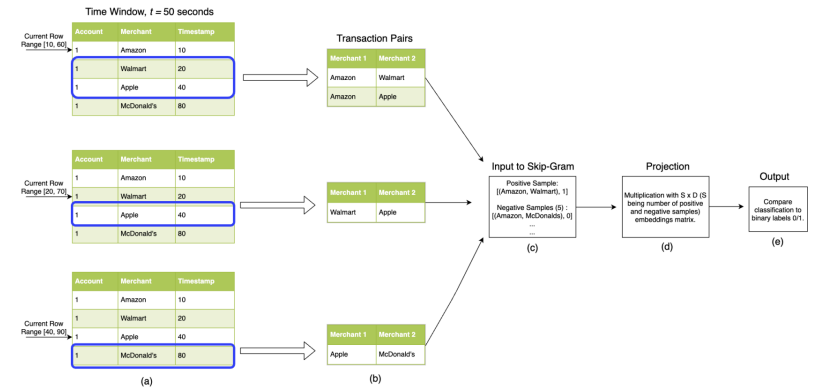


Fig. 1. Model pipeline, from data pre-processing to training with Skip-gram. Using stringent time windows and pairs of transaction pairs (example time window, $t = 50$ seconds) allows for creating meaningful embeddings on graphs with millions of unique entities.

Adidas	Delta Air Lines	West Elm
Nike	United Airlines	Anthropologie
Reebok	American Airlines	Crate&Barrel
Timberland	Frontier Airlines	Pottery Barn
KFC	Starbucks	Gap
Taco Bell	Panera Bread	Banana Republic
Little Caesars	Dunkin Donuts	Ann Taylor Loft
Burger King	Jamba Juice	J Crew

TABLE III
MERCHANT NEAREST NEIGHBORS DERIVED FROM THE BRAND-LEVEL MERCHANT GRAPH. FOR EACH QUERY MERCHANT VECTOR (IN BOLD), THE TOP THREE NEAREST NEIGHBOR MERCHANT VECTORS ARE SHOWN.

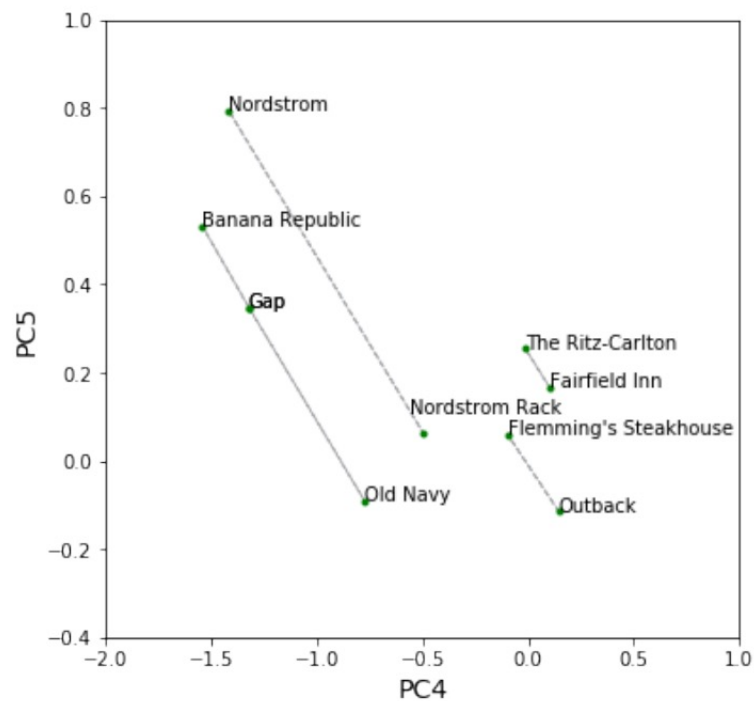


Fig. 4. Merchant embeddings tend to encode typical price point of goods. Merchants from multiple industries visualized in the subspace spanned by the fourth and fifth principal components. Pairs of merchants are joined by dotted lines, with each pair containing a high-end merchant and a more affordable counterpart. Across disparate industries, the relationship between high-end offerings and affordable offerings is embedded in a consistent direction of this space.

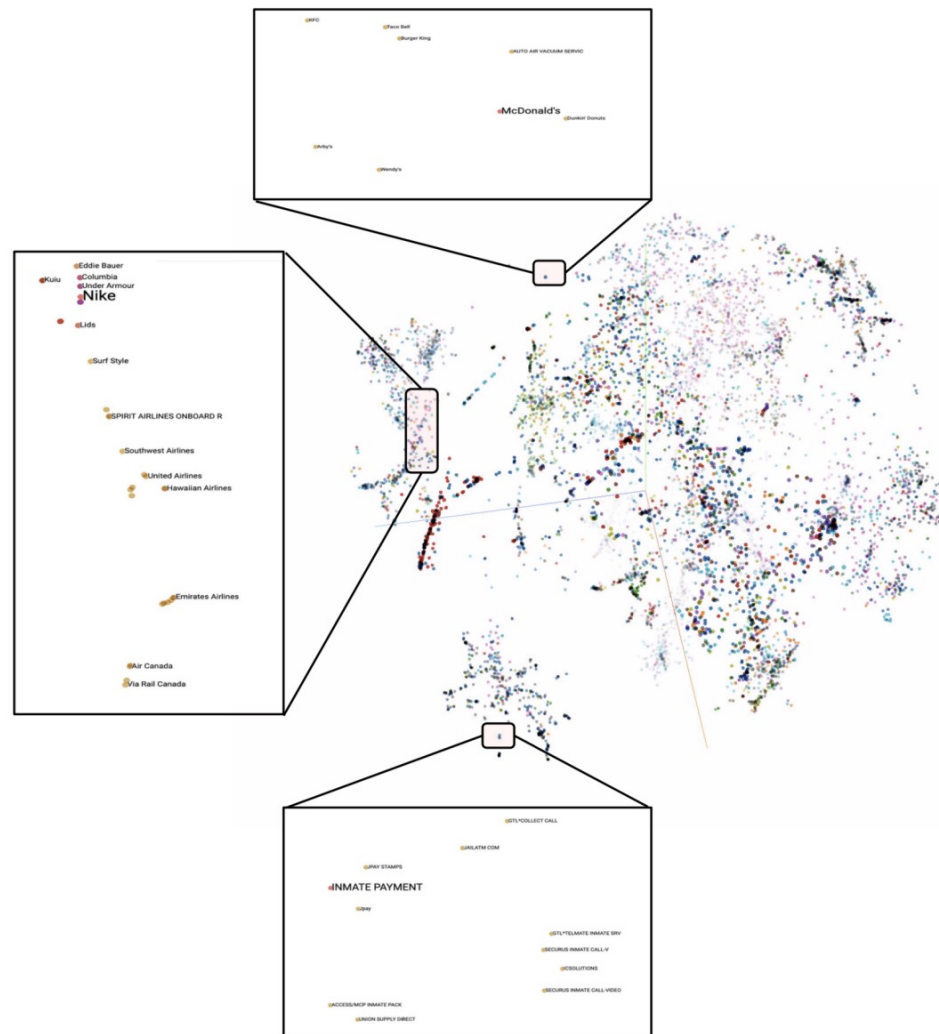


Fig. 3. Example clusters found in a two dimensional t-SNE projection of brand-level embeddings. Best viewed digitally.

C. Application to Fraud Detection

We also assess the quality of our raw merchant embeddings by evaluating them in a transfer learning task involving transaction fraud detection. Results on these experiments are reported relative to a baseline model (the details of which we omit here) and are quantified by the area under the precision-recall curve (AUpr). In Table IV, we see that directly using the trained embeddings from the raw merchant graph yields a roughly 1.0% improvement in fraud classification AUpr when these embeddings are included as additional features to the

model. One downside of this approach is the large expansion of the feature space in order to include the embeddings. To overcome this, we additionally tested a model whereby we added a trainable MLP that took as input the embedding for a transaction’s merchant and predicted a binary output for fraud. This ancillary model, once trained, can then be used to output a single fraud score per transaction, conveying only the information contained in the merchant embedding space that is useful for fraud detection. This single score is then passed into the base model as an additional feature and yields a 5.2% boost in efficacy. Merchants that engage in fraudulent transactions typically engage with similar vendors over all transaction pairs. As our model’s objective is to encode transactional behavior within each merchant’s embedding representation, we attribute an increase in classification accuracy to capturing semantically meaningful features that allow the classifier to better identify likely-fraudulent merchants.

The Total Economic Impact™ Of Experian Ascend Platform

Business Benefits And Cost Savings Enabled By Experian Ascend Platform

[Experian Ascend Platform™](#) is a cloud-based analytics, decisioning, and fraud platform that leverages robust data and artificial intelligence to improve credit and fraud decisions.¹ The Experian Ascend Platform provides advanced analytics and automates the review process, which ultimately enables more precise credit decisions, higher operational efficiency, improved financial performance, and reduced risk.

- **Default cost reduction of 20%.** Experian Ascend Platform enhances the composite organization's credit risk modeling and eliminates manual errors in fraud checks. This enables the composite to reject fraudulent applications more accurately and reduce credit risks, leading to lower default rates. This benefit is worth \$3.2 million to the composite.

Understanding Structured Financial Data with LLMs: A Case Study on Fraud Detection

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Abstract

Detecting fraud in financial transactions typically relies on tabular models that demand heavy feature engineering to handle high-dimensional data and offer limited interpretability, making it difficult for humans to understand predictions. Large Language Models (LLMs), in contrast, can produce human-readable explanations and facilitate feature analysis, potentially reducing the manual workload of fraud analysts and informing system refinements. However, they perform poorly when applied directly to tabular fraud detection due to the difficulty of reasoning over many features, the extreme class imbalance, and the absence of contextual information. To bridge this gap, we introduce FinFRE-RAG, a two-stage approach that applies importance-guided feature reduction to serialize a compact subset of numeric/categorical attributes into natural language and performs retrieval-augmented in-context learning over label-aware, instance-level exemplars. Across four public fraud datasets and three families of open-weight LLMs, FinFRE-RAG substantially improves F1/MCC over direct prompting and is competitive with strong tabular baselines in several settings. Although these LLMs still lag behind specialized classifiers, they narrow the performance gap and provide interpretable rationales, highlighting their value as assistive tools in fraud analysis.

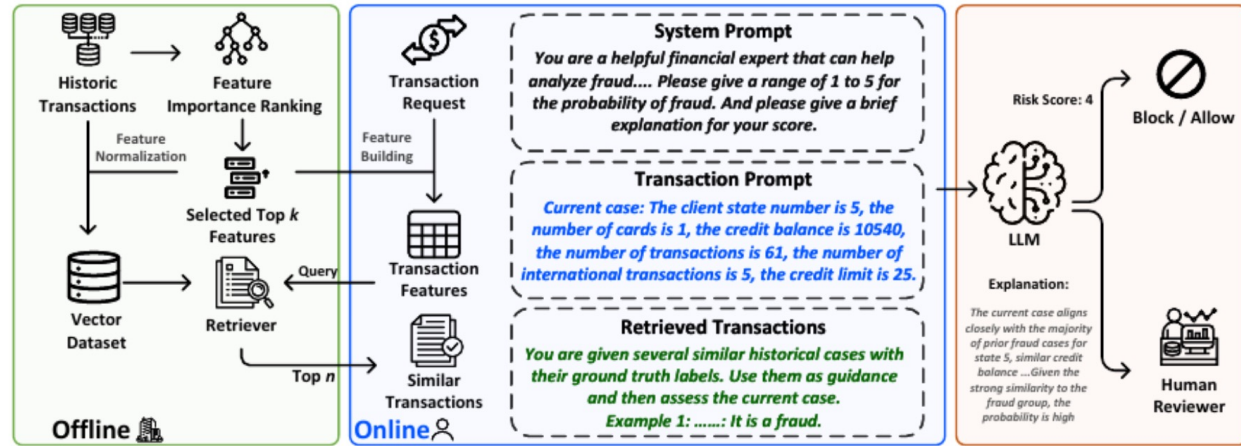


Figure 1: Overall architecture of the FinFRE-RAG framework.

Transaction Description of a Fraud Case

You are a helpful financial expert.....Please give a range of 1 to 5 for the probability of fraud.....The transaction has attributes: TransactionDT: 10372086, TransactionAmt: 12.205, card1: 2801, C13: 0.000, card2: 130.000, C14: 0.000, addr1: missing, P_emaildomain: gmail.com, card6: credit, C1: 3.000.

GPT-OSS-20B

Score: 2 – The transaction amount ($\approx \$12$) is low, which is typical for many legitimate purchases. The email domain is a common, reputable provider (gmail.com). The card type is a standard credit card (card6 = credit). The only notable red flag is the missing addr1 value, which can sometimes indicate incomplete data but on its own is not a strong fraud indicator. Overall, the limited risk factors suggest a relatively low probability of fraud, though the missing address keeps it from being the absolute lowest score.

GPT-OSS-20B with FinFRE-RAG

Score: 4 – The transaction **shares many key fraud-signatures seen** in the training set: it uses a Gmail address, a credit card, a small amount ($\sim \$12$), and a high C1 value (3.0). Most Gmail-credit transactions with $C1 \geq 2$ in the examples were labeled fraud (e.g., examples 1, 4, 9, 11, 16). The only notable differences are the relatively low card1 value (2801) and the missing address field, but these alone have not been strong enough to override the other strong fraud indicators. Thus the probability of fraud is high, though not absolute.

Figure 4: Example responses from GPT-OSS-20B without and with FinFRE-RAG. With in-context learning on similar transactions, it learns to use patterns between different transactions to identify potential fraud.

	AUD		IEEE-CIS				PAYSIM			
	Prec.	Rec.	F1	MCC	Prec.	Rec.	F1	MCC	Prec.	Rec.
FinFRE-RAG	0.27	0.32	0.15	0.71	0.31	0.45	0.21	0.28	0.10	0.85
GPT-OSS-120B	0.01	0.06	0.01	0.64	0.14	0.08	0.08	0.77	0.04	1.00
+ FinFRE-RAG	0.44	0.46	0.33	0.64	0.53	0.51	0.63	0.46	0.66	0.85
Random Forest	0.85	0.85	0.92	0.78	0.52	0.52	0.39	0.82	0.79	1.00
XGBoost	0.89	0.89	0.92	0.86	0.48	0.50	0.32	0.92	0.68	0.92
TabM	0.85	0.85	0.92	0.79	0.66	0.65	0.71	0.62	0.92	0.85

Table 1: Performance comparison of baseline models and FinFRE-RAG on fraud datasets. Each block reports F1, MCC, Precision, and Recall; higher scores indicate better performance.

Graphical Methods

Graphical Methods

- Fraud is **relational** not just **transactional**
- Network analysis enables understanding of broader set of counterparties and relationships
- Represent through “nodes” capturing individuals/firms and “edges” to capture transactions and relationships
 - In contrast to “relational databases” which are valuable for indexing and storing data, but do not capture these hidden relationships
- Link Analysis: Connecting transactions to discover suspicious connections
- Community detection: identifying clusters of suspicious activities

Graph Neural Networks for Financial Fraud Detection: A Review

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Abstract The landscape of financial transactions has grown increasingly complex due to the expansion of global economic integration and advancements in information technology. This complexity poses greater challenges in detecting and managing financial fraud. This review explores the role of Graph Neural Networks (GNNs) in addressing these challenges by proposing a unified framework that categorizes existing GNN methodologies applied to financial fraud detection. Specifically, by examining a series of detailed research questions, this review delves into the suitability of GNNs for financial fraud detection, their deployment in real-world scenarios, and the design considerations that enhance their effectiveness. This review reveals that GNNs are exceptionally adept at capturing complex relational patterns and dynamics within financial networks, significantly outperforming traditional fraud detection methods. Unlike previous surveys that often overlook the specific potentials of GNNs or address them only superficially, our review provides a comprehensive, structured analysis, distinctly focusing on the multifaceted applications and deployments of GNNs in financial fraud detection. This review not only highlights the potential of GNNs to improve fraud detection mechanisms but also identifies current gaps and outlines future research directions to enhance their deployment in financial systems. Through a structured review of over 100 studies, this review paper contributes to the understanding of GNN applications in financial fraud detection, offering insights into their adaptability and potential integration strategies.

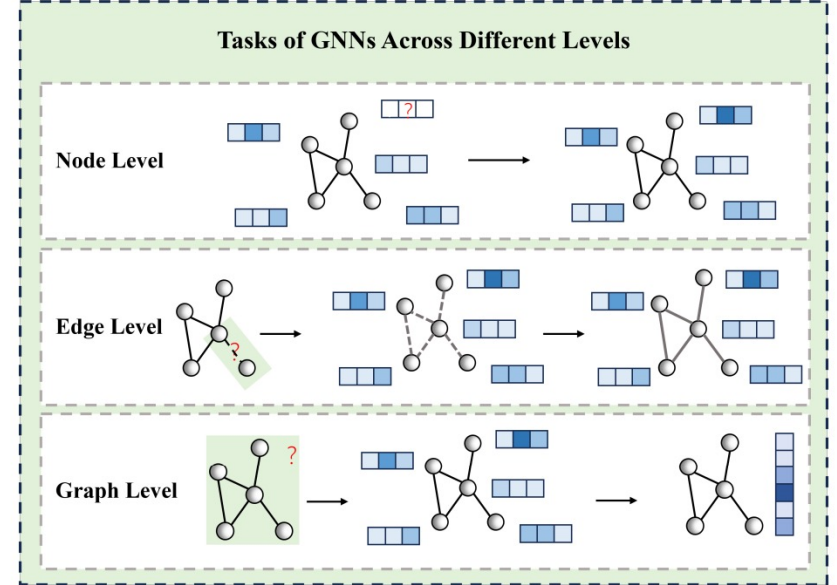


Fig. 2 The tasks of GNNs at different levels. For node level, the label of each node is determined by its features and neighbors. For edge level, the label of each edge is determined by the features of source & target node. For graph level, the property is determined by the features of all nodes or edges.

Graphical Methods of Fraud Detection

To address the requirements of fraud detection within the financial sector, constructing financial-oriented graphs is essential for uncovering *abnormal transaction patterns*, *unusual behavioral interactions*, and *potential fraud clusters*. This subsection introduces various methods for constructing graphs based on financial data, aiming to facilitate the detection and prediction of fraudulent behavior:

Transaction Graph Construction. Transaction data is pivotal for revealing fraudulent activities. Creating a transaction network (graph), with nodes representing individual accounts or entities (such as individuals, companies) and edges representing transactions (e.g., transfers, payments) [74, 75], enables the effective identification of abnormal transaction patterns [76]. For example, frequent small transactions or large funds flows within a short period may indicate fraudulent activity [77]. Analyzing the network's topology, such as node centrality and community divisions, can help identify fraud rings or key fraudulent accounts.

Relationship-Based Graph Construction. Beyond direct transaction data, the relationships between financial entities

are crucial for fraud detection [78]. Constructing complex network (graph) that include various entities (like individuals, companies, bank accounts) and their relationships (such as ownership, control relations) can reveal the cooperation networks behind complex fraudulent activities like money laundering or insider trading [79]. For example, Mao and others utilize the inter-enterprise relationships to establish a network [80].

Behavioral and Device Pattern Graph. Constructing graphs based on user behaviors and device usage patterns (e.g., logins, queries, transaction requests, device types, operating systems, and device IDs) can reveal the similarities or abnormal patterns between different accounts [81–83]. By connecting accounts whose behavior and device usage similarity exceeds a certain threshold, it's possible to uncover networks of accounts potentially operated by fraudsters. This includes identifying users who switch devices in patterns that deviate from typical user behavior, or who use devices known to be associated with fraudulent activities [58]. Such graphs significantly enhance machine learning models' ability to recognize atypical behavior and device usage patterns, thereby predicting and identifying potential fraud more effectively [24].

Information Flow Graph Construction. Malicious actors often exploit misleading information for market manipulation. Constructing information flow graphs [84] by analyzing the relationship between public information (e.g., news, announcements, social media discussions) and financial market activities can help identify patterns of information manipu-

Application: Estimating Dark Shipping

Charting the Uncharted: The (Un)Intended Consequences of Oil Sanctions and Dark Shipping

Abstract

We examine the rise of dark shipping – oil tankers disabling AIS transceivers to evade detection – amid Western sanctions on Iran, Syria, North Korea, Venezuela, and Russia. Using a machine learning-based ship clustering model, we track dark-shipped crude oil trade flows worldwide and detect unauthorized ship-to-ship transfers. From 2017 to 2023, dark ships transported an estimated 7.8 million metric tons of crude oil monthly – 43% of global seaborne crude exports – with China absorbing 15%. These sanctioned flows offset recorded declines in global oil exports but create distinct economic shifts. The U.S., a net oil exporter, faces lower oil prices but benefits from cheaper Chinese imports, driving deflationary growth. The EU, a net importer, contends with rising energy costs yet gains from Chinese demand, fueling inflationary expansion. China, leveraging discounted oil, boosts industrial output, propagating global economic shocks. Our findings expose dark shipping's central role in reshaping oil markets and macroeconomic dynamics.

JEL-Codes: C320, C380, E320, Q430, R400.

Keywords: dark shipping, oil sanction, satellite data, clustering analysis, LP.

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Dark Shipping Clustering Identification

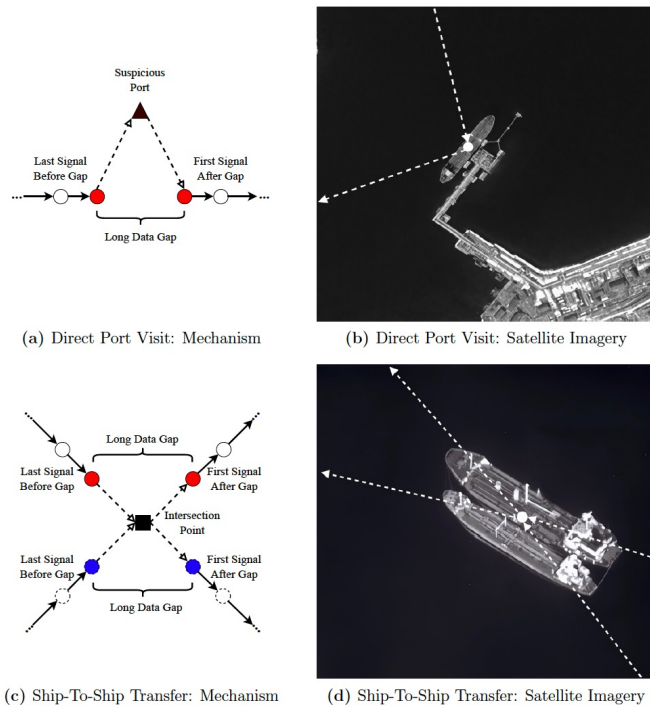


Figure 1: Deceptive Shipping Practices During Prolonged AIS Data Gaps

Notes. Panel 1a illustrates a vessel returning to a nearby port in a sanctioned country during a prolonged AIS data gap. Panel 1b presents satellite imagery of such a visit by the identified dark ship *Roma* (IMO: 9182291), with dashed white arrows indicating the vessel's trajectory during the data gap. Panel 1c shows a ship-to-ship transfer in which both vessels disable their AIS transceivers, creating overlapping data gaps. The intersection point is estimated using the last known coordinates and headings before the AIS signals disappear. Panel 1d provides satellite imagery of a dark ship-to-ship transfer involving two identified dark ships, *Abyss* (IMO: 9157765) and *Shanaye Queen* (IMO: 9242118), with dashed white arrows representing the trajectories of both vessels.

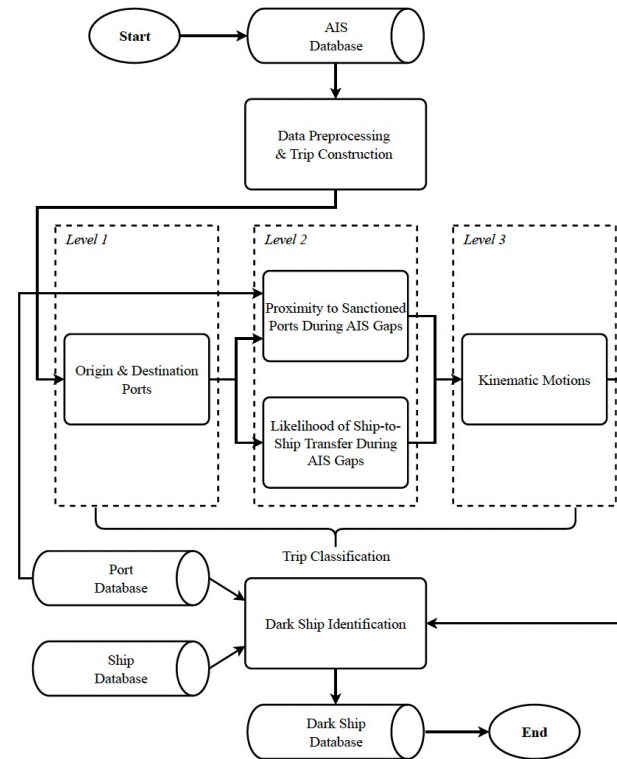


Figure A.1: Methodology Framework of the Multi-Attribute Ship Clustering Model

Geographical Concentration of Dark Ships

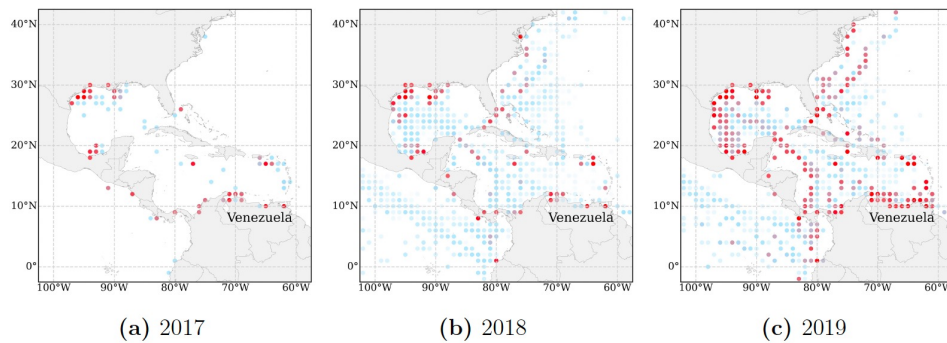


Figure 4: Geographical Concentration of Dark Shipping Activities Near Venezuela: 2017–2019

Notes. Geographical concentration of dark shipping activities near Venezuela from 2017 to 2019. Each colored point represents a geographical square (with 1-degree variations in longitude and latitude) that contains at least one last recorded data point before a vessel disables its AIS transceiver. The chromatic intensity of each point reflects the sum of port-based trip suspicion scores associated with these last recorded data points. The port-based trip suspicion score estimates the likelihood that a vessel traveled to a nearby sanctioned-country port during its AIS gap.

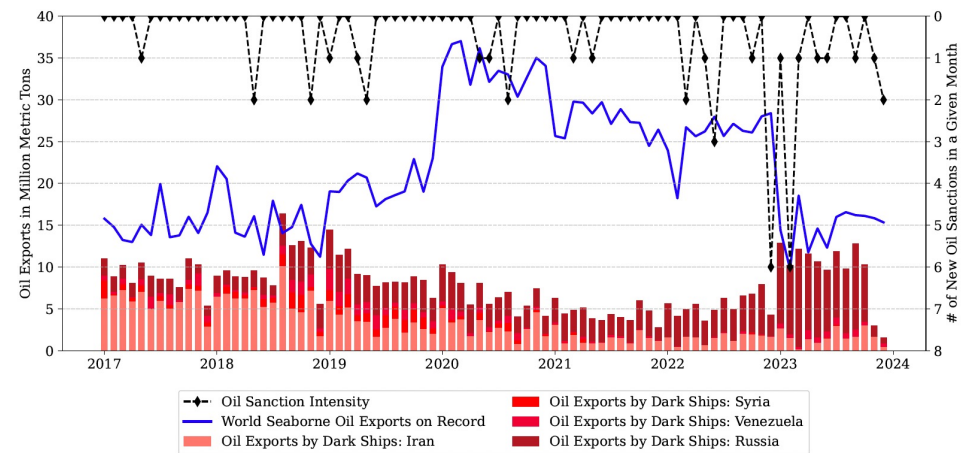


Figure 9: Oil Exports of Sanctioned Countries by Dark Ships, World Seaborne Oil Exports on Record, and Oil Sanction Intensity: 2017M1–2023M12

Application: Pig Butchering Fraud

How Do Crypto Flows Finance Slavery? The Economics of Pig Butchering

JOHN M. GRIFFIN and KEVIN MEI*

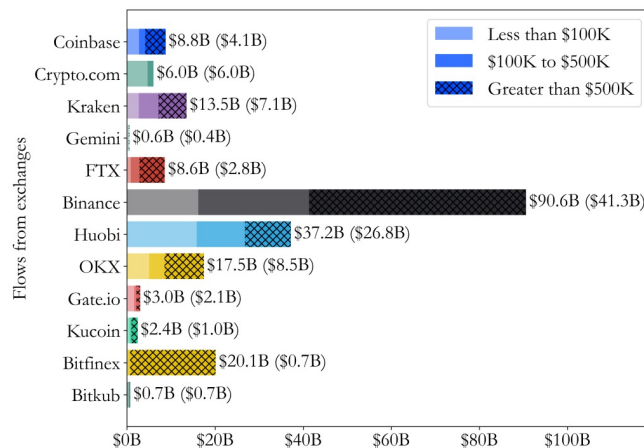
Abstract

Through blockchain addresses, we trace crypto flows and uncover methods commonly used by scammers to obfuscate their activities. The perpetrators interact freely with major crypto exchanges, sending over 32,000 small trust-building inducement payments annually to exchanges commonly used by U.S. and European investors. Funds exit the Ethereum network in large quantities, mostly in Tether, through less transparent but large exchanges. Criminal enterprises pay approximately 33 basis points in transaction fees and moved approximately \$27.8 billion annually into suspicious exchange deposit accounts between 2021-2023. Our findings highlight how many actors in the “reputable” crypto industry facilitate criminal capital flows.

Random social media or text messages attempting to develop an online relationship are now commonplace. In a subset of cases, friendly online relationships slowly transition into “pig butchering scams” or *sha zhu pan*, which often bleed lonely, sick, and distressed victims of their life savings. Though varied in nature, the origin of these scams is often even darker, as an estimated 220,000 individuals are forcibly held in compounds in Southeast Asia to operate these scams, constituting modern-day slavery. Many investigative reports, government briefs, and documentaries describe how this scheme lures victims globally with elaborate schemes of friendship or romance orchestrated by individuals confined in Southeast Asian compounds.¹ This paper examines how these criminal organizations are financed through cryptocurrencies. How do criminal networks use crypto to move victim funds? Where does capital enter? Where do the funds exit? What obfuscation methods are employed? How pervasive is this activity? How can it be stopped?

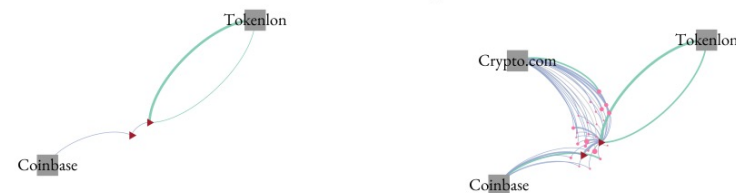
[illegible]

This figure plots the originating exchange of funds that later flow into our traced scammer network addresses, with the horizontal axis representing the total flow for each exchange on the vertical axis. This considers any addresses found within the forward or backward trace that had less than 2,000 transactions, transacts with less than five tokens, only uses included functions, and is not an otherwise already identified exchange. Transaction flows are grouped according to whether the transactions are less than \$100,000, between \$100,000 and \$500,000, or greater than \$500,000; respectively these groups are represented by lighter and darker shades of the corresponding exchange color. We present the sum of each at the top of each bar, and the sum of transactions less than \$500,000 is presented in parentheses.

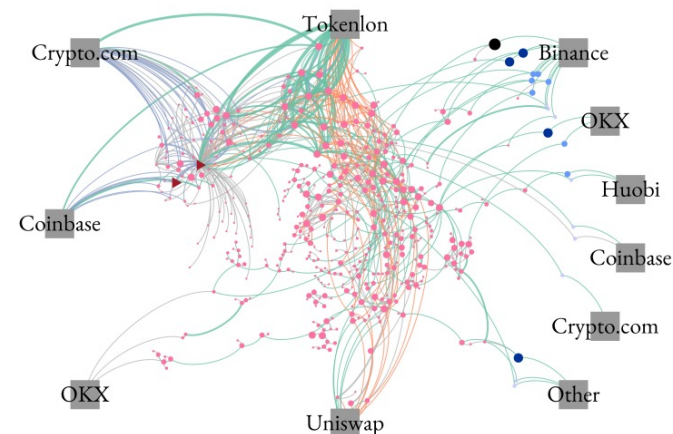


This figure shows the flow of funds that can be attributed to a victim report. The report includes two addresses plotted in dark red in Panel A. The victim sent funds to the left red node and were later transferred to the right red node, which swapped the funds into Tether. Panel B and C plot the trace originating from the right red address or collection node, with Panel B plotting the other inflows into the collection node (i.e., right red node), and Panel C plotting the trace of the swapped funds leaving the collection node. Edge colors correspond to the most commonly transacted token between nodes (see bottom left legend). All other conventions follow Figure 1.

(b) Other flow into the reported collection point



(c) Trace paths from the reported collection point



Nodes

- ETH
- USDT
- USDC
- DAI

Reported addresses
Address found in trace

Reported addresses
Address found in trace
Deposit addresses (<\$10K, <\$100K, <\$1M, >\$1M)

Explainable Machine Learning in Deployment

Umang Bhatt^{1,2,3,4}, Alice Xiang², Shubham Sharma⁵, Adrian Weller^{3,4,6}, Ankur Taly⁷, Yunhan Jia⁸,
Joydeep Ghosh^{5,9}, Ruchir Puri¹⁰, José M. F. Moura¹, Peter Eckersley²

¹Carnegie Mellon University, ²Partnership on AI, ³University of Cambridge, ⁴Leverhulme CFI, ⁵University of Texas at Austin, ⁶The Alan Turing Institute, ⁷Fiddler Labs, ⁸Baidu, ⁹CognitiveScale, ¹⁰IBM Research

ABSTRACT

Explainable machine learning offers the potential to provide stakeholders with insights into model behavior by using various methods such as feature importance scores, counterfactual explanations, or influential training data. Yet there is little understanding of how organizations use these methods in practice. This study explores how organizations view and use explainability for stakeholder consumption. We find that, currently, the majority of deployments are not for end users affected by the model but rather for machine learning engineers, who use explainability to debug the model itself. There is thus a gap between explainability in practice and the goal of transparency, since explanations primarily serve internal stakeholders rather than external ones. Our study synthesizes the limitations of current explainability techniques that hamper their use for end users. To facilitate end user interaction, we develop a framework for establishing clear goals for explainability. We end by discussing concerns raised regarding explainability.

Ways to Address Explainability

4.2.2 Shapley Values in Practice. Organization A works with financial institutions and helps explain models for credit risk analysis. To integrate into the existing ML workflow of these institutions, Organization A proceeds as follows. They let data scientists train a model to the desired accuracy. Note that Organization A focuses mostly on models trained on tabular data, though they are beginning to venture into unstructured data (i.e., language and images). During model validation, risk analysts conduct stress tests before deploying the model to loan officers and other decision-makers. After decision-makers vet the model outputs as a sanity check and decide whether or not to override the model output, Organization A generates Shapley value explanations.

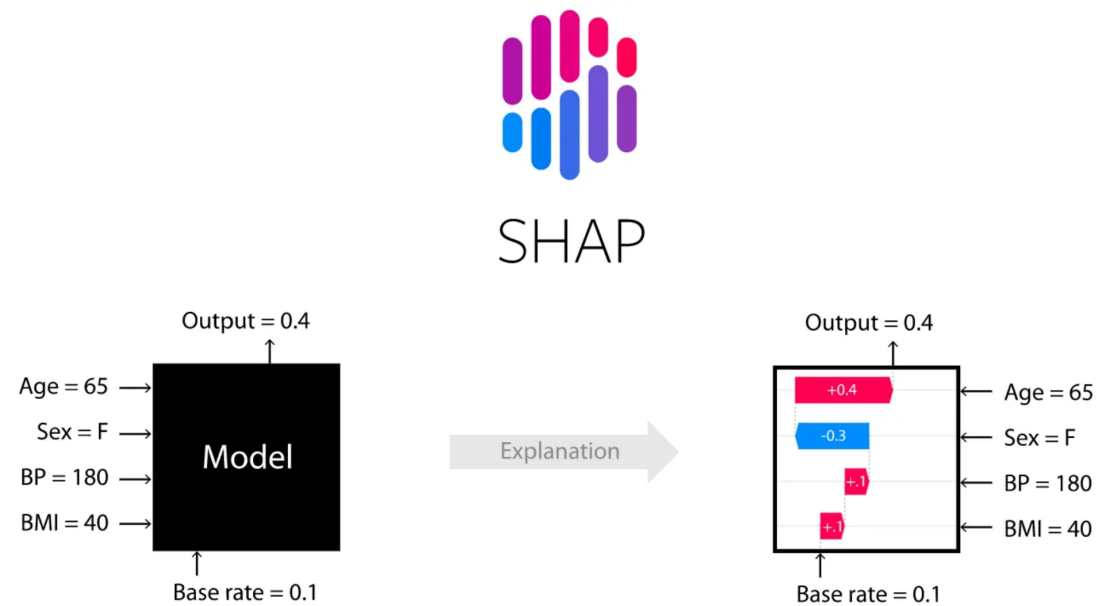
Before launching the model, risk analysts are asked to review the Shapley value explanations to ensure that the model exhibits expected behavior (i.e., the model uses the same features that a human would for the same task). Notably, the customer support team at these institutions can also use these explanations to provide individuals information about what went into the decision-making process for their loan approval or cash distribution decision. They are shown the percentage contribution to the model output (the positive ℓ_1 norm of the Shapley value explanation along with the sign of contribution). This means that the explanation would be along the lines of, “55% of the decision was decided by your age, which positively correlated with the predicted outcome.”

4.3.1 Formulation. Counterfactual explanations are points close to the input for which the decision of the classifier changes. For example, for a person who was rejected for a loan by a ML model, a counterfactual explanation would possibly suggest: “Had your income been greater by \$5000, the loan would have been granted.”

5.2.4 Beyond Deep Learning. Though deep learning has gained popularity in recent years, many organizations still use classical ML techniques (e.g., logistic regression, support vector machines), likely due to a need for simpler, more interpretable models [52].

Many in the explainability community have focused on interpreting black-box deep learning models, even though practitioners

Shapley Values



SHAP - Header (Source: github.com/slundberg/shap)

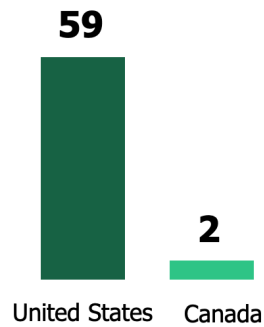
Regulatory and Compliance

High Cost of Financial Compliance

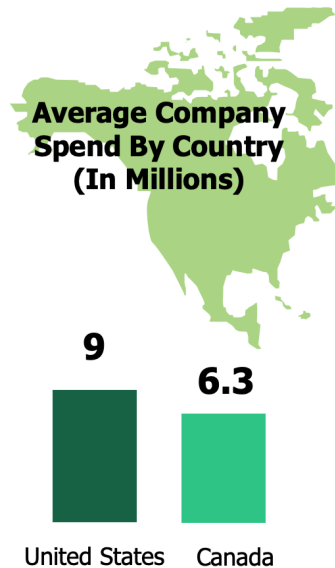
**Total Cost Of Financial
Crime Compliance**



**Total Cost Of Financial
Crime Compliance By
Country (In Billions)**



**Average Company
Spend By Country
(In Millions)**



Note: The total annual cost of financial crime compliance is calculated using the number of financial institutions in the surveyed markets and survey data regarding financial crime costs. A spend amount is generated for each region by multiplying its average reported total cost of financial crime compliance operations by the number of financial institutions in that region.

Big Four Hiring AI Auditors Even Before ChatGPT!

952

A. Fedyk et al.

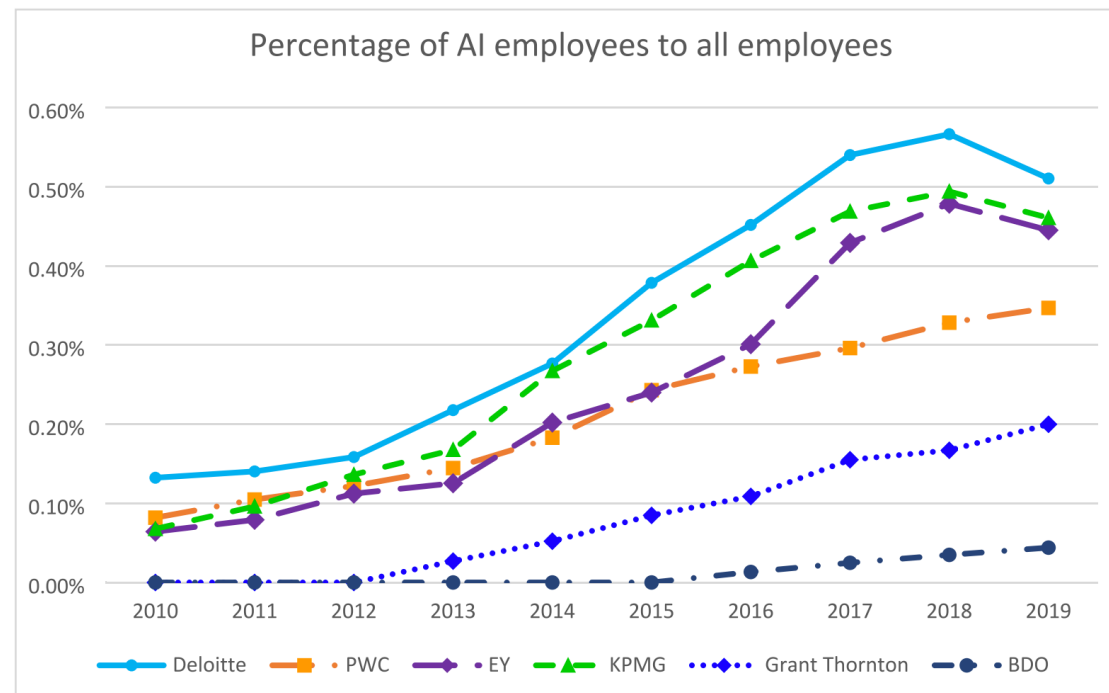
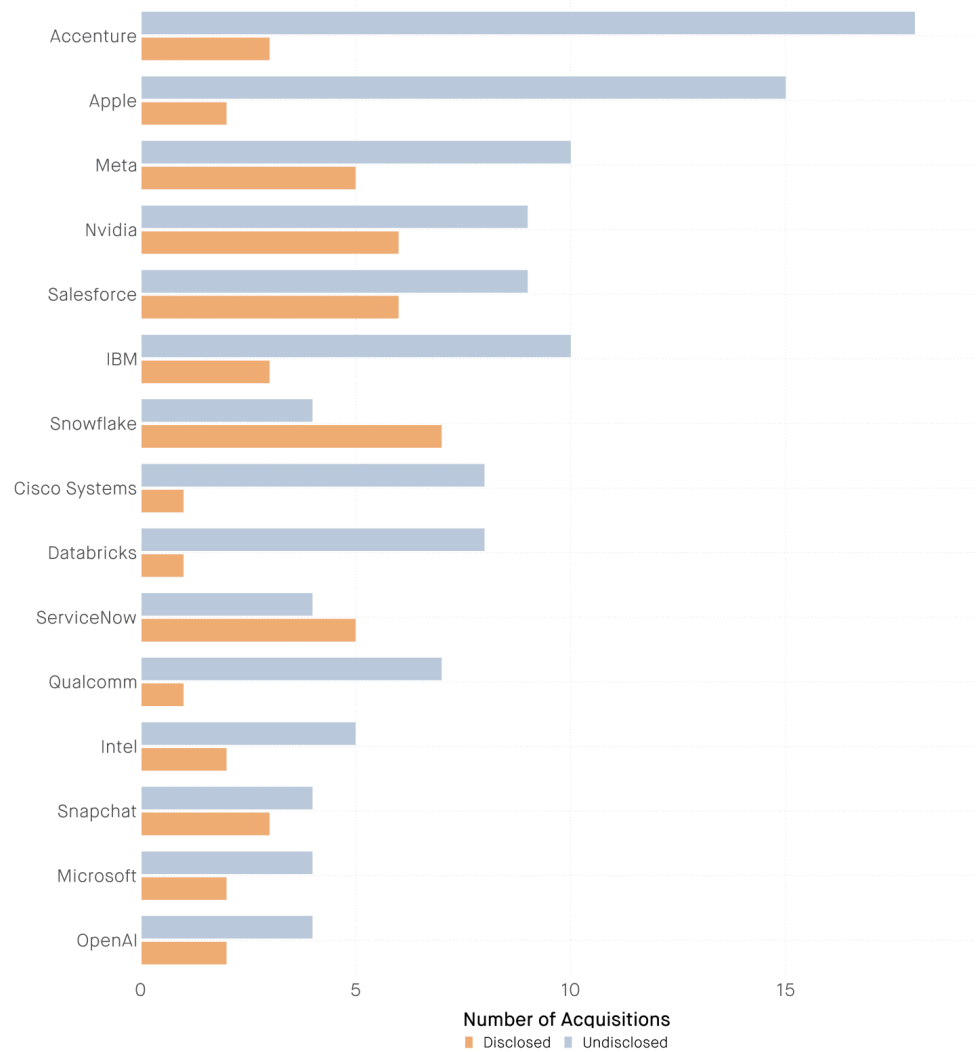


Fig. 1 AI Employees in the Six Largest Audit Firms

Top AI Acquirers : Disclosed vs Undisclosed Deals

Most acquisitions across all buyers have undisclosed valuations, 2020-2025

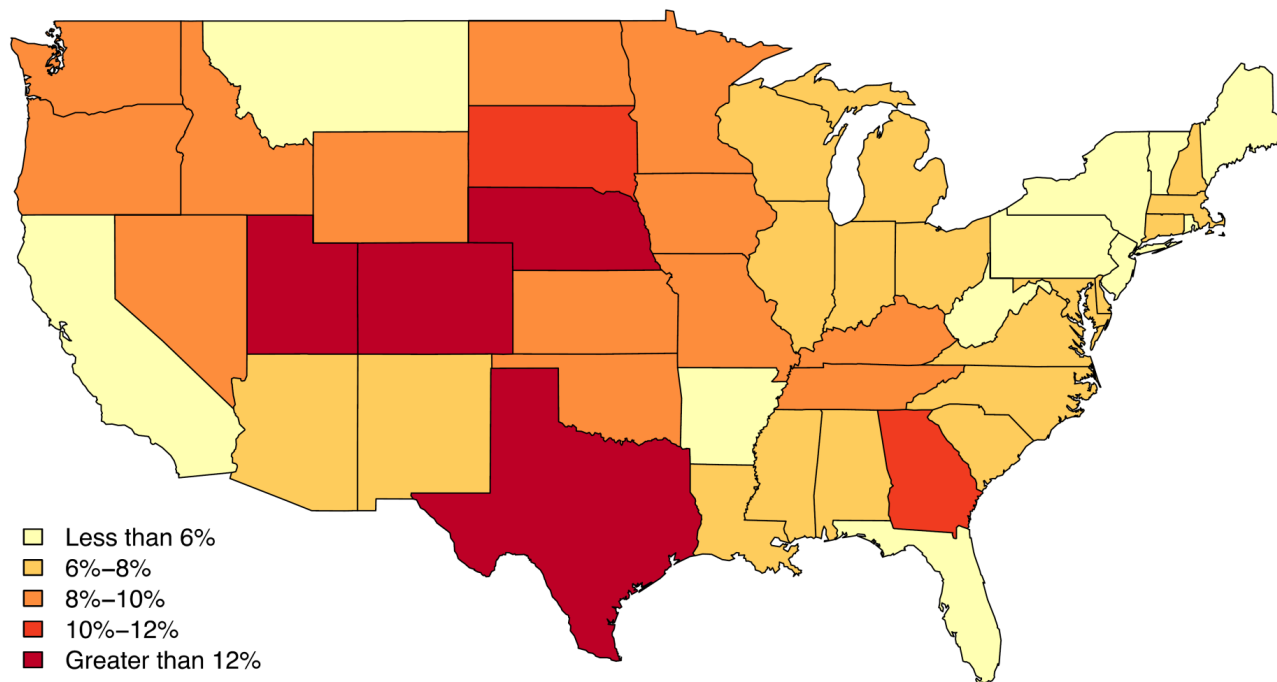


Regulatory Compliance Issues are Rising

- Firms pre-emptively spend a lot to address financial crimes
- Penalties for failing to comply with AML/KYC, sanctions, customer due diligence etc. totaled \$6.6b in 2023
- Firms face reputational risk
 - Danske Bank money laundering scandal; \$2b in fines, CEO resigned etc.
- Operational burden of dealing with compliance issues
- Dealing with multiple jurisdictions is a hassle
- Ultimately leads to loss in credit, and shifts to private markets/crypto/etc.

Misrepresentation of presence of junior lien

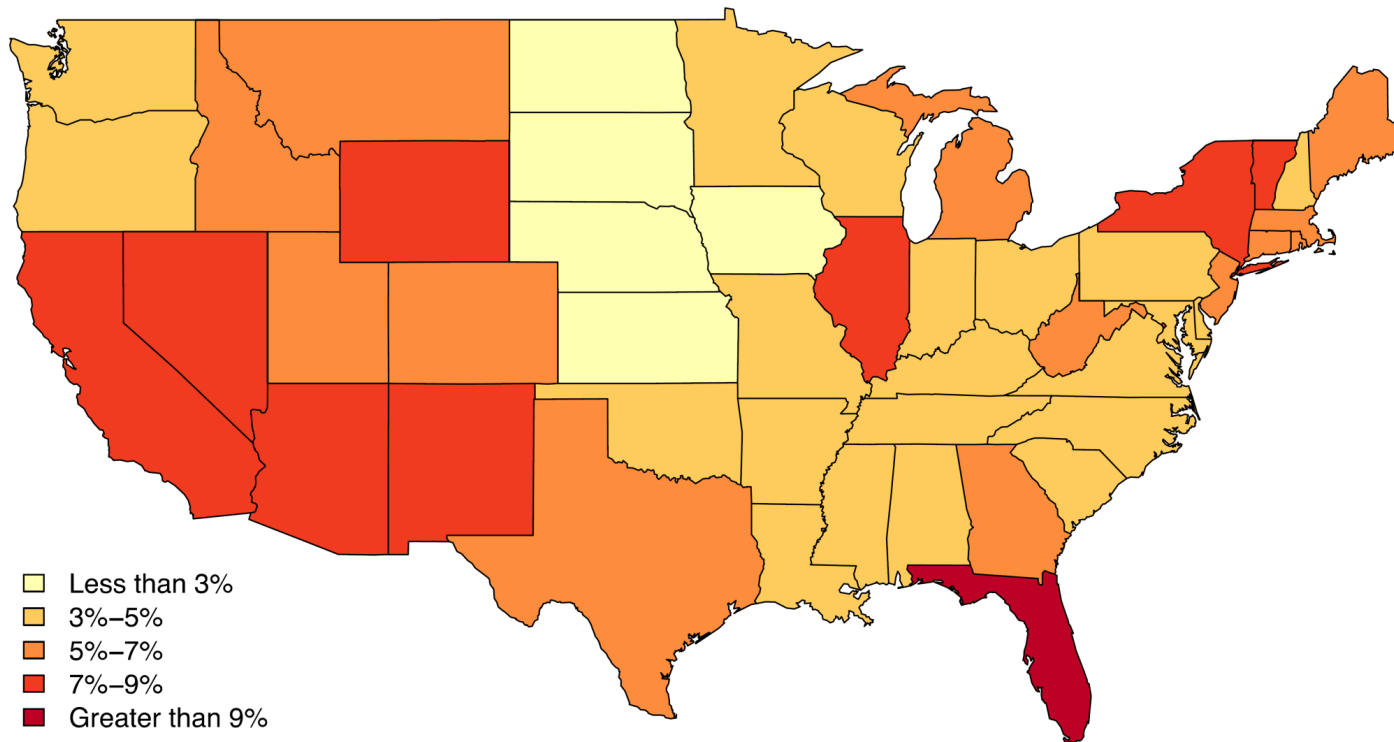
Figure 1B: Fraction of Loans Reported as Having No Second Lien that Misreport Second Lien



Piskorski, Seru and Witkin (2013)

Misrepresentation of occupancy status

Figure 1A: Fraction of Loans Reported as Owner-Occupied that Misreport Occupancy Status

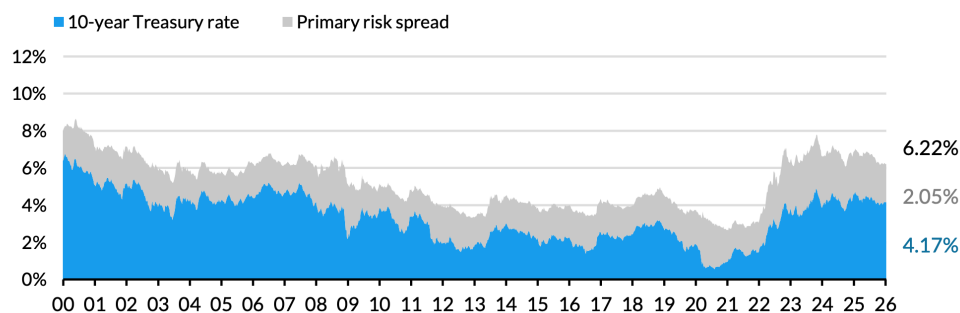


Piskorski, Seru and Witkin (2013)

Why Hasn't Technology Improved Mortgage Pricing?

30-Year Fixed Mortgage Commitment Rate

By 10-year Treasury and primary risk spread



Source: Board of Governors of the Federal Reserve System and the Freddie Mac Primary Mortgage Market Survey and the Urban Institute.

Notes: Data as of January 9, 2025. The primary risk spread is the difference between the 30-year fixed mortgage rate and the 10-year Treasury note rate.

Government Litigation Risk and the Decline in Low-Income Mortgage Lending

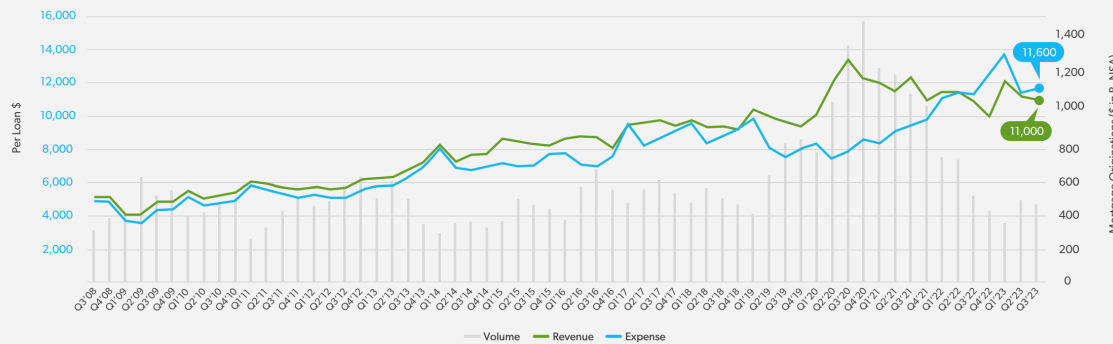
W. Scott Frame, Kristopher Gerardi, Erik J. Mayer, Billy Y. Xu, and Lawrence Chengzhi Zhao

Working Paper 2024-6
July 2024

Abstract: We study the effect of Department of Justice lawsuits in the 2010s against large lenders for alleged fraud in the Federal Housing Administration (FHA) mortgage insurance program. The suits led to more than \$5 billion in settlements and caused targeted banks and their peers to precipitously exit the FHA market. Difference-in-differences and triple differences tests exploiting geographic variation in exposure to exiting banks show a 20 percent reduction in FHA lending in heavily exposed areas. This reduction was not associated with improved underwriting standards or lower default rates. Large banks' FHA exit has significantly reduced low-income households' overall access to mortgage credit.

Compliance and Operational Costs

EXHIBIT 1: AVERAGE LOAN PRODUCTION REVENUE, EXPENSE (\$ PER LOAN) & LOAN VOLUME



Note: Cost and revenue production results include retail/consumer direct only lender data.
Source: Freddie Mac, Moody's, MBA's Quarterly Mortgage Bankers Performance Report; www.mba.org/performance-report



1 Operational Costs are Rising

Average origination costs have increased by ~\$3,000 per loan in the past three years. To potentially mitigate the increase in costs and gain efficiency, many lending institutions are leaning into technology adoption, with more digitally mature organizations showcasing higher benefits/efficiencies. In fact, the top cost-effective companies originate loans nearly half the cost of the industry average and ~2.4 times less than the bottom 25% performers.



3 Digitization is the Path Forward

In interviews with executives, it's clear that mortgage firms recognize the benefits of a fully digitized mortgage process and believe it can help them save up to 40% in costs. While the leading tech adopters are interested in artificial intelligence (AI), blockchain and predictive analytics, the path forward varies depending on where they are in their digital journey and how they should be responsibly incorporating these tools into their processes.

Housing in America

Comradely capitalism

How America accidentally nationalised its mortgage market

Aug 20th 2016 | From the print edition

The machine blew up in 2006-10 for a host of reasons, the most important of which was wild and sometimes fraudulent underwriting. There was a run on mortgage bonds and on the firms that issued or owned them. There have since been three big changes.

The trouble with Gosplan

First, banks have partially withdrawn from the mortgage game after facing swathes of new rules and \$110 billion of fines for misconduct. They still own mortgage-backed bonds and they still make home loans to wealthy folk, which they keep on their balance-sheets. But with the exception of Wells Fargo

they are less keen on writing riskier loans in their branches and feeding them to securitisers. New, independent firms like Quicken Loans and Freedom Mortgage have filled the gap. They originate roughly half of all new mortgages.

Decline in Lending to Firms by Banks

The Decline of Big-Bank Lending to Small Business: Dynamic Impacts on Local Credit and Labor Markets

Brian S. Chen, Samuel G. Hanson, and Jeremy C. Stein

NBER Working Paper No. 23843

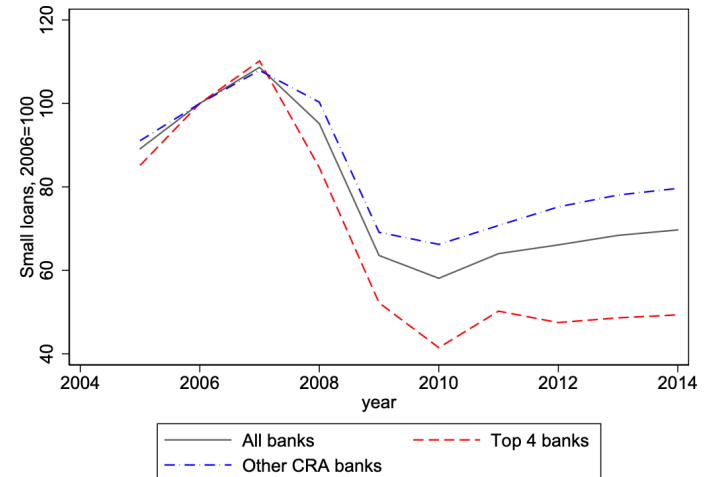
September 2017

JEL No. G01,G21,G23

ABSTRACT

Small business lending by the four largest banks fell sharply relative to others in 2008 and remained depressed through 2014. We explore the dynamic adjustment process following this credit supply shock. In counties where the largest banks had a high market share, the aggregate flow of small business credit fell, interest rates rose, fewer businesses expanded, unemployment rose, and wages fell from 2006 to 2010. While the flow of credit recovered after 2010 as other lenders slowly filled the void, interest rates remain elevated. Although unemployment returns to normal by 2014, the effect on wages persists in these areas.

Figure 1: Annual originations of small business loans by Top 4 U.S. banks vs. all other banks



AI in Compliance Stack

- Potentially, AI can be used to speed up regulatory and compliance paperwork to address vulnerabilities and speed up credit access
- Drastically improved capabilities relative to previous technology in both information retrieval as well as regulatory document paper trail creation
- However, the cost of errors is typically **high** which is a major bottleneck to wider adoption
- And broader rollout of Gen AI technologies risks creating new vulnerabilities

Cybersecurity Risks

- Deepfakes and social engineering
 - Very convincing impersonation of customers, employees, and executives for fraud
 - Dramatically lowered costs of phishing and other attacks
- Injection attacks: inputting malicious content

AI-Driven Fraud and Impersonation: New Face of Financial Crime

and Impersonation: The New Face of Financial Crime

FRI | JUL 18, 2025 | 6:12 AM PDT

In early 2024, an employee at a Hong Kong firm joined what appeared to be a routine video meeting with her chief financial officer and colleagues. By the end of the call, she had authorized \$25 million in transfers to overseas accounts. Weeks later came the shocking truth: every "colleague" on that call, including the CFO, was a **sophisticated AI-generated deepfake**. This incident, among others, heralds a new era of fraud in which artificial intelligence enables criminals to impersonate trusted individuals with uncanny realism.

Adversarial Attacks on AI Models

- Evasion: understand what model is targeting and avoid that
- Data Poisoning: submitting false data to skew model's classifications
- Prompt Injection: "Ignore all previous instructions..."
- Data privacy and leakage
- Synthetic Identity creation
- AI arms race: advancing capabilities for both fraudsters and financial institutions